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EVEN END SEMESTER EXAMINATION - 2025

Name of the Course: B.Tech. (ME)

Semester: VIII

Name of the Paper: Automatic Control

Paper Code: TRA 801

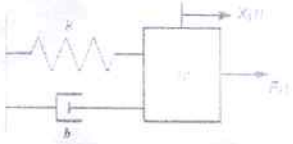
Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Describe the open-loop and closed-loop control systems with figures.	CO1
(b)	Plot and write the expressions for the following signals: step, ramp, and sinusoidal.	
(c)	Find the solution $x(t)$ of the following differential equation: $\ddot{x} + 3\dot{x} + 2x = 0 ; \text{ where } x(0) = p \text{ and } \dot{x}(0) = q$	
Q2	(20 marks)	
(a)	What are the poles and zeros of a system's transfer function. Also, describe the effect of poles location on the plane of real and imaginary axes (s-plane) in order to predict the stability.	CO2
(b)	What is the Routh's stability criterion? Discuss about its necessary and sufficient conditions.	
(c)	Comment about the stability of a system using Routh's stability criterion, whose characteristic equation is given by the following polynomial: $s^4 + 2s^3 + 3s^2 + 5s + 4 = 0$	
Q3	(20 marks)	
(a)	Describe the following transient-response characteristics of a control system with a neat sketch: (i) Delay time, (ii) Rise time, and (iii) Maximum overshoot.	CO3
(b)	Derive the expression for the unit-step response of a first-order system.	
(c)	Derive the expression for the unit-step response of a second-order undamped system.	
Q4	(20 marks)	
(a)	What do you understand by the modern control theory? Write the standard state space equations for a linear dynamic system.	CO4

(b)	Derive the transfer function of the mass-spring-dashpot system as shown in the figure below. Also, comment about the stability of the system if, $m=2$ kg, $b=1$ N-s/m, and $k=3$ N/m. 	
(c)	What do you understand by root locus? Write the main steps to plot the root locus of a system.	
Q5	(20 marks)	
(a)	What is a bode plot? Write the different steps to draw the bode plot.	CO5 & CO6
(b)	Describe the frequency response analysis of a system with mathematical expression.	
(c)	Describe the Proportional-Integral-Derivative (PID) control action with governing equations.	